Design of MgH_2-Ni-2D Nanosheets Composites for Enhancing Hydrogen Storage/Release property Through Nickel Complex Precursor and Graphene/HB

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Abstract: This research aims to improve the kinetics and hydrogen release/storage capacity of magnesium hydride MgH_2 by adding Ni catalyst. It is reported that the electronegativity of the surrounding element of nickel-based compound influences hydrogen release/storage characteristics of MgH_2-Ni composite. However, the mechanism of why this occurs, to the best of our knowledge, is not widely elucidated. Also, Mg needs to be negative to induce quicker hydrogen release. We wish to uncover the reason behind this mechanism and optimise the charge state of Ni and Mg to improve hydrogen storage/release kinetics by comparing the influence of several nickel complexes on de/hydrogenation of MgH_2-Ni . We will ball mill a mix of MgH_2 and several nickel complex salts precursors to form Ni–MgH_2 nanocomposite. Furthermore, a two-dimensional nano-sheet such as hydrogen boride (HB) or graphene will be added later to the composite to prevent 1) agglomeration of MgH_2 nanocrystals and degradation effects created by ball milling and 2) improve the cycle performance of Mg/MgH_2. The performance of ball-milled MgH_2–Ni and MgH_2–Ni–HB/graphene composite will then be compared. During PBL II, we experiment with Mg–Nickel complex salts composites first. We first start with investigating the hydrogen release properties of ball-milled MgH_2–$Ni(acac)_2$ under two different ball-milled conditions: "intensive method" and "gently ball-milled method". This is to optimise ball milling conditions for future nickel complexes for promoting hydrogen release/storage.

Keywords: Hydrogen storage, Solid-state hydrogen storage, MgH_2, Magnesium hydride, Nickel catalyst, Ni, Nickel-complex salt precursor, ball milling, hydrogen boride, graphene, 2D layered nanosheet, nano-encapsulation, nano-encapsulated composite.

1 Introduction

Hydrogen is widely regarded as a clean and sustainable energy carrier with the potential to replace fossil fuels and contribute to carbon neutrality [1–3]. Among hydrogen storage methods, solid-state storage using metal hydrides offers high gravimetric and volumetric hydrogen densities, improved safety, and economic advantages [1,4]. MgH_2 is one of the most promising hydrogen storage materials because of its abundance, high hydrogen storage capacity (7.6 wt.%), thermodynamic stability, and safe handling [4,5]. However, its practical application is limited by slow hydrogen desorption kinetics, poor cycling stability, and the need for high hydrogen release temperatures (>300 °C) [1,4].

To overcome these limitations, transition metal catalysts, particularly Ni, and two-dimensional (2D) materials such as graphene or hydrogen boride (HB) have attracted considerable attention [1,4]. Nickel catalysts accelerate hydrogen dissociation and recombination, lower the activation energy for hydrogen absorption/desorption, and improve reaction kinetics, while reducing catalyst particle size enhances catalytic activity [4]. Meanwhile, 2D materials help suppress grain agglomeration during cycling, improving kinetics, cycle stability, and hydrogen storage performance [1,4]. Consequently, combining a Ni catalyst precursor with a 2D nanomaterial is expected to enhance the dehydrogenation kinetics and long-term cycling stability of MgH_2 while maintaining a high hydrogen storage capacity (>7 wt.%) [4].

It is reported that the electronegativity of the element bonded to Ni in a nickel-based compound that is doped into MgH_2 affects the hydrogen release and storage kinetics and performance of MgH_2. [16] The break of the bond between the metal and H ion is easier as the metal has lower electronegativity. [19] As Ni binds with the low-electronegativity element, the stability of the Ni–H bond decreases. [16, 19] Nickel complex consists of different elements, and we theorise each element and the surrounding ligand around Ni has diverse effects on the kinetics of hydrogen release/ storage of MgH_2. For

quicker hydrogen H-release, Mg should be more negative: electronegativity in bonding with Mg form a role. Conversely, Mg should more positive if we want to induce a quicker H- absorption.[15] Also, the ligand in the nickel complex affect the reaction pathways. The acetylacetonate in $Ni(acac)_2$ affects the reaction pathways of dehydrogenation in MgH_2+$Ni(acac)_2$ and induces the hydrogen pump "Mg_2Ni". [4] Therefore, the ligand around Ni in a nickel complex plays an important factor.

In this research, we will dope several nickel complexes as a Ni catalyst precursor into MgH_2 by ball milling. Then, we measure dehydrogenation/hydrogenation of MgH_2, and consider the ligands around nickel in affecting the charge state of Ni and Mg which affects Mg-H and Ni-H and thus the dehydrogenation/ hydrogenation kinetics of MgH_2.

This research therefore aims to reduce the hydrogen release temperature, improve hydrogen absorption/desorption kinetics, enhance cycling stability, and maintain the high hydrogen storage capacity of MgH_2 by doping MgH_2 with nickel complexes and, in a later stage, incorporating HB or graphene. Ball milling will be used both to reduce particle size and to uniformly disperse the catalyst within the MgH_2 matrix [6]. Previous studies have shown that ball milling significantly improves hydrogen storage performance by refining particle size and facilitating catalyst incorporation. For example, ball milling reduced the hydrogen release temperature of TiH_2 by approximately 150 °C, while similar approaches lowered the activation energy of $LiAlH_4$ and $LiBH_4$ through catalyst addition [3,6,7].

Our hypothesis is that ball-milled MgH_2–Ni composites will exhibit lower hydrogen desorption temperatures, faster dehydrogenation kinetics, improved cycling stability, and better hydrogen storage performance than pure ball-milled MgH_2. The surrounding ligand in a nickel complex will affect the multivalence state of Ni, charge of Mg, bonding of Ni-H and affect hydrogen release/storage. Furthermore, the addition of graphene or HB is expected to prevent agglomeration of MgH_2 and Ni particles, thereby further enhancing cycle stability [1,4].

The study will first compare the dehydrogenation behavior of MgH_2 doped with several nickel complexes, including $Ni(acac)_2$, $NiCl_2 \cdot 6H_2O$, $Ni(OH)_2$, nickel phthalocyanine (NiPc), and nickelocene (Cp_2Ni), to investigate how the ligand environment around Ni influences hydrogen release. (Figure 1) $Ni(acac)_2$ has been selected as the initial model catalyst precursor to optimize the ball-milling conditions due to the high performance of hydrogen release and storage of MgH_2+$Ni(acac)_2$ [4]. The experimental workflow includes pre-characterization, ball milling, post-milling characterization, hydrogen desorption measurements using TG–DTA and temperature-programmed desorption (TPD), and post-desorption characterization, with X-ray diffraction (XRD) used to evaluate crystallinity. Two ball-milling conditions ("intensive" and "gentle") will be compared with pure ball-milled MgH_2 to identify reproducible processing conditions. After optimisation, the same workflow will be applied to the remaining nickel complexes, with the most promising catalyst precursor subsequently evaluated for hydrogen storage using pressure–composition–temperature (PCT) measurements. Finally, graphene or HB will be incorporated into the best-performing MgH_2–Ni composite to further improve hydrogen storage performance and cycling stability [1,4,6].

We begin our research by investigating and seeking to compare dehydrogenation kinetics and characteristics of MgH_2+nickel complexes. Figure 1 shows example of the hydrogen release profile we want to obtain from Temperature Programmed Desorption(TPD) to prove our hypothesis: temperature decrease of hydrogen release of MgH_2- nickel complex as compared to that of pure MgH_2.

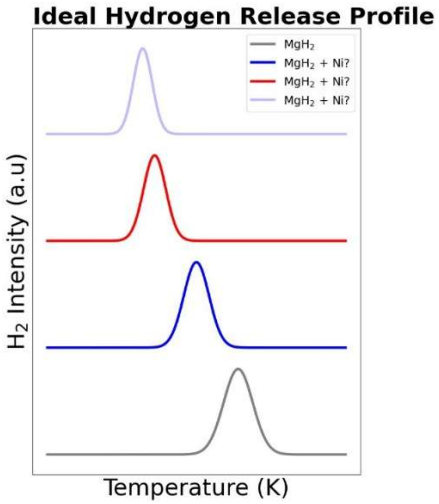


Figure 1 An example of hydrogen release profile of MgH_2+ nickel complex salt compared to pure MgH2 we want to obtain to prove our hypothesis

2. Related Works: Nickel Complexes

2.1 Metal hydrides and chemisorption [13]

MgH2 belongs to binary metal hydrides. Metal hydrides bond with hydrogen through chemisorption. Chemisorption-based hydrogen storage relies on the dissociation of hydrogen molecules into atoms, which then form strong chemical bonds with the storage materials. Due to these strong interactions, chemisorption materials can achieve higher hydrogen densities than physisorption-based systems.

Metallic hydrides form when metals or alloys react with hydrogen, offering a safer and more efficient storage option than compressed or liquid hydrogen. These materials exhibit significantly higher hydrogen densities and operate under moderate temperatures and pressures.

Formation of metal hydrides

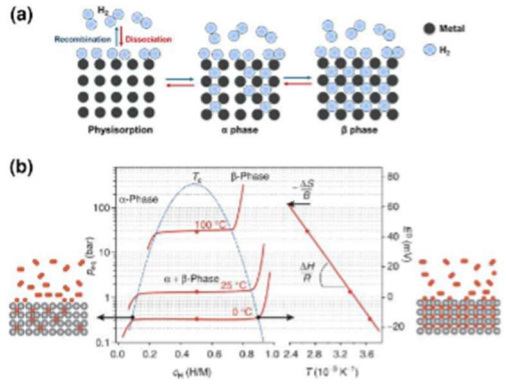


Figure 2 a) Schematic diagram of the formation of metal hydride. b) Pressure–composition–temperature (PCT) plot (left) and a van't Hoff curve (right) collected from LaNi5. The experimental value of the enthalpy of formation (ΔH) can be determined from the slope of the van't Hoff plot. Elsevier 2009 [14, 13]

MgH_2 is a binary metal hydride that stores hydrogen through chemisorption, where hydrogen molecules first dissociate into atoms before forming strong chemical bonds with the metal lattice. These strong metal–hydrogen interactions enable metal hydrides to achieve higher hydrogen storage densities than physisorption-based materials while providing a safer alternative to compressed or liquid hydrogen under moderate operating conditions.

The formation of metal hydrides begins with the dissociation of hydrogen molecules, followed by diffusion of hydrogen atoms into the metal lattice (Figure 2). During hydrogen absorption, the host metal first forms a hydrogen solid solution (α-phase). As hydrogen concentration increases, nucleation and growth of the hydride (β-phase) occur. The coexistence of the α- and β-phases produces a plateau region in the pressure–composition isotherm, corresponding to the equilibrium pressure and reversible hydrogen storage capacity (Figure 2b). Hydride formation is accompanied by a 20–30% volumetric expansion, allowing large amounts of hydrogen to be stored through strong chemisorption bonds within the metal lattice.

Binary hydrides

Metal hydrides are generally classified into binary hydrides and intermetallic hydrides. Binary hydrides, particularly those based on lightweight elements such as Li, Be, Mg, and Al, offer high gravimetric hydrogen storage capacities. However, their practical application is limited by slow hydrogen absorption/desorption kinetics and high thermodynamic stability.

Among these materials, MgH_2 is one of the most promising hydrogen storage media because of its high volumetric (110 kg m^{-3}) and gravimetric (7.6 wt.%) hydrogen storage capacities, as well as its reversible hydrogenation/dehydrogenation under moderate conditions (350–400°C, 35 bar). Despite these advantages, MgH_2 suffers from slow hydrogen desorption kinetics due to its high dehydrogenation enthalpy and low thermal conductivity, motivating extensive research to improve its performance.

Several approaches have been developed to enhance MgH_2, including nanostructuring, alloying, and catalytic doping. Nanostructuring increases surface energy and grain boundaries, improving hydrogen diffusion and reaction rates. Huot et al. first demonstrated this approach in 1999 by using high-energy ball milling to reduce MgH_2 particle size to approximately 7.5 μm, resulting in a fivefold improvement in hydrogen absorption and desorption kinetics.

Alloying MgH_2 with elements such as Co, Cd, Al, Si, Ni, and Fe forms intermetallic compounds with lower thermodynamic stability, improving hydrogen storage kinetics and reducing operating temperatures. For example, Zaluska's group (1999) developed nanocrystalline Mg alloys containing transition-metal catalysts, while Mg_2Ni exhibited a lower formation enthalpy (-64 kJ mol^{-1}) than pure MgH_2 (-74 kJ mol^{-1}). Aluminum-based Mg alloys also showed improved kinetic and thermodynamic properties. However, alloying reduces gravimetric hydrogen capacity because the added metals increase the composite's weight. For instance, a nanostructured Mg–Ti alloy showed improved formation enthalpy (-45 ± 3 kJ mol^{-1}) but its hydrogen storage capacity decreased to 4.0 wt.%, compared with 7.6 wt.% for pure MgH_2. Similar reductions were reported for Mg_2Ni (3.6 wt.%), Mg_2Cu (2.7 wt.%), and Mg_2Fe (5.4 wt.%). Consequently, catalytic metals are typically added in small amounts (1–5 wt.%) to minimize capacity loss. [95,97–99]

Catalytic doping is another effective strategy for improving MgH_2 performance by reducing the activation energy for hydrogen absorption and desorption. Various catalysts, including transition metals, oxides, nitrides, and halides, have been shown to enhance reversible hydrogen storage kinetics. Catalysts improve performance by lowering activation energy, promoting hydrogen chemisorption, accelerating hydrogen diffusion into the Mg matrix, and providing nucleation sites for hydride formation. [13]

2.2 Role of Ni-based compounds in improving hydrogen release/storage properties of MgH2 [16]

Each nickel complex consists of different elements, and each element has diverse effects on the kinetics of hydrogen release/ storage of MgH_2. For quicker hydrogen H-release, Mg should be less negative: electronegativity in bonding with Mg form a role. [15]
Zhang et al. (2017) systematically investigated the catalytic effects of different Ni-based compounds on the dehydrogenation performance of MgH_2 by comparing MgH_2–X composites (X = Ni_3C, Ni_3N, NiO, and Ni_2P) [15]. The nanoscale Ni-based catalysts were synthesized using various chemical methods and subsequently incorporated into MgH_2 through mechanical ball milling. The dehydrogenation behavior was evaluated using isothermal dehydrogenation, temperature-programmed desorption (TPD), and differential scanning calorimetry (DSC).

Building on their earlier work, Zhang et al. demonstrated that the anion species in Ni-based compounds strongly influences catalytic activity, as electron transfer from the anion to Ni creates an electron-rich nickel surface that enhances hydrogen desorption [17,18]. Ni_3C, Ni_3N, and Ni_2P exhibit particularly high catalytic activity because of their non-compensated electronic structures, while NiO is also a well-established hydrogen storage catalyst.

Among the catalysts tested, the dehydrogenation performance ranked as:
MgH_2–Ni_3C > MgH_2–Ni_3N > MgH_2–NiO > MgH_2–Ni_2P [15].
MgH_2–Ni_3C showed the best performance, with the lowest onset dehydrogenation temperature (160 °C), followed by MgH_2–Ni_3N (180 °C), MgH_2–NiO (205 °C), and MgH_2–Ni_2P (248 °C). At 300 °C for 20 minutes, the composites released 6.2, 4.9, 4.1, and 3.5 wt.% H_2, respectively. Correspondingly, their activation energies were reduced to 97.8, 100.0, 119.7, and 132.5 kJ mol^{-1}, indicating that lower activation energy contributes to faster hydrogen desorption kinetics. All composites also exhibited lower dehydrogenation enthalpies than pure MgH_2, reflecting improved thermodynamic properties, although hydrogen storage capacities decreased slightly because of catalyst addition. MgH_2–Ni_3C and MgH_2–Ni_3N retained the highest hydrogen capacities (6.5 wt.%), whereas MgH_2–Ni_2P had the lowest (6.1 wt.%). [16]

Because MgH_2–Ni_3C demonstrated the best overall

performance, Zhang et al. further optimized the Ni_3C content by preparing MgH_2–x wt.% Ni_3C (x = 1, 3, 5, 7, and 10) composites [15]. All composites exhibited an onset dehydrogenation temperature of approximately 160 °C, while increasing Ni_3C content shifted the peak desorption temperature to lower values. Among the compositions studied, MgH_2–5 wt.% Ni_3C showed the best overall hydrogen storage performance, achieving 5.7 wt.% hydrogen desorption upon heating to 300 °C and 6.2 wt.% hydrogen release within 20 minutes at 300 °C, while also exhibiting the fastest desorption kinetics. These results indicate that 5 wt.% Ni_3C is the optimum catalyst loading, providing the best balance between hydrogen storage capacity and dehydrogenation performance [16].

Dehydrogenation Process of MgH2-Nickel based MgH2-X (X=Ni3C, Ni3N, NiO and Ni2P) composites

The dehydrogenation process of the MgH2-X (X =Ni_3C, Ni_3N, NiO and Ni_2P) composites is a gas-solid reaction, whose reaction rate is affected significantly by the gas-solid interface. Figure 3 shows a schematic of the effect of Ni-based compounds on the dehydrogenation of MgH2.

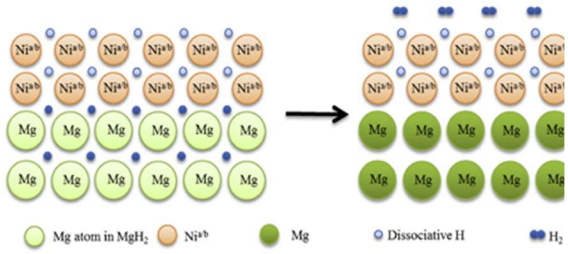


Figure 3 The schematic diagram of the dehydrogenation of MgH2-X (X =Ni3C, Ni3N, NiO and Ni2P) composites

$Ni^{x/y}$ stands for the multivalences of the Ni-based compound. Notably, the $Ni^{x/y}$ can gain electrons more easily than Mg2+, and thus act as an electron transfer medium during dehydrogenation process.

According to [19], the point of contact of Mg/MgH2 with a transition metal TM-based catalyst by ball-milling can be schematically drawn as in Figure 4.

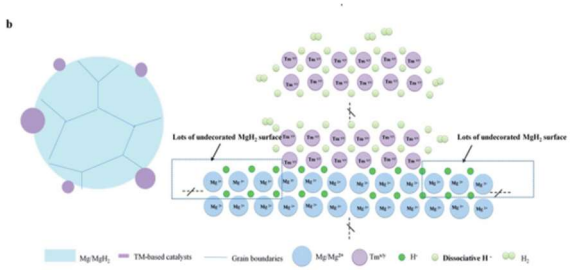


Figure 4 A schematic diagram of the ball-milled Mg-Transition Metal (TM) catalyst. From Figure 9b in [19]

In this case, only some of the H ions on the surface of MgH2 can interact with the TM $^{x/y}$ ions. Due to the d orbital characteristics of a TM, $TM_{x/y}$ ions can lose or gain electrons more easily than Mg^{2+}, thus acting as an electron transfer medium during the hy-/dehydrogenation process.

The dehydrogenation of Mg-based composites proceeds as the following: [16]

Step 1: H^- between the Mg^{2+} and Ni^{xy} tends to diffuse to Ni^{xy} and form the Ni–H bonds.

Step 2: H^- diffuses across the Ni^{xy} layer and arrives on the surfaces.

Step 3: The Ni–H bonds break.

Based on the dehydrogenation process, the hydrogen desorption relates to the formation of Ni–H bonds, transformation of H, and the breaking of Ni–H bonds. Moreover, the breaking of Ni–H bonds is the control step. It is reported that the break of the bond between the metal and H ion is easier as the metal has lower electronegativity. [19] As the Ni binds with the low-electronegativity element, the stability of the Ni–H bond decreases. The electronegativity of C, N, O, and P is 2.55, 3.04, 3.44, and 1.64, respectively. Therefore, the dehydrogenation performance of the Mg-based composites ranks as MgH_2–Ni_3C, MgH_2–Ni_3N, and MgH_2–NiO. P has the lowest electronegativity; however, the dehydrogenation mechanism of MgH_2–Ni_2P composites is an exception due to the formation of Mg_2Ni and Mg_3P_2.

2.3 Hydrogen desorption of MgH2-Ni3N[18]

Zhang *et al.* investigated the effect of nitrogen-containing nickel catalysts by synthesizing a core–shell Ni_3N@NC catalyst, consisting of a ~16 nm Ni_3N core coated with a ~2 nm N-doped carbon shell, and incorporating it into MgH_2–x wt.% Ni_3N@NC (x = 3, 5,

10, and 15) composites through ball milling [15]. Among the compositions studied, MgH_2–5 wt.% Ni_3N@NC exhibited the best hydrogen storage performance. Its onset hydrogen desorption temperature decreased significantly to 175 °C, compared with 335 °C for pure MgH_2, and it released 6.0 wt.% H_2 within 20 minutes at 325 °C, whereas pure MgH_2 released only 0.2 wt.% H_2 under the same conditions. Hydrogen desorption was found to follow a two-dimensional nucleation and growth mechanism [15].
The enhanced catalytic activity was attributed to the electron-donating effect of nitrogen, which increases the electron density on the nickel surface, destabilizing the Mg–H bonds and thereby improving dehydrogenation performance [15]. In addition, the N-doped carbon shell suppresses grain agglomeration during dehydrogenation and cycling while providing additional active sites and hydrogen diffusion pathways, leading to improved dehydrogenation kinetics and reversibility of MgH_2 [15].

2.4 MgH2-NiO and "hydrogen pump" Mg2Ni/ MgNiH4 [20]

Tian *et al.* prepared porous nanosheet NiO via hydrothermal synthesis of $Ni(OH)_2$ followed by calcination, and incorporated 5 wt.% NiO into MgH_2 by ball milling to improve its hydrogen storage performance [19]. The MgH_2–5 wt.% NiO composite exhibited a significantly reduced onset dehydrogenation temperature of 210 °C, which is 114 °C lower than that of ball-milled MgH_2. It released 6.4 wt.% H_2 within 10 minutes at 300 °C and began hydrogen uptake at room temperature. The composite also demonstrated excellent cycling stability, retaining 96.2% of its hydrogen storage capacity after 40 cycles. Additionally, the NiO catalyst reduced the hydrogen desorption activation energy from 133.69 ± 12.55 kJ mol^{-1} to 81.47 ± 7.48 kJ mol^{-1}. These improvements were attributed to the in situ formation of Mg_2Ni/Mg_2NiH_4, which act as a "hydrogen pump" by weakening the Mg–H bonds and thereby enhancing the hydrogen storage properties of MgH_2 [19].
The "hydrogen pump" generated by the in situ formed Mg2Ni/Mg2Ni4 during the hydrogen absorption and desorption process is as of the following: [21]

1. Upon hydrogen absorption,Mg2Ni preferentially adsorbs hydrogen, to form Mg_2NiH_4 and transfers hydrogen to the Mg interface, promoting the hydrogenation of Mg.

$$MgNi + H_2 \rightarrow MgNiH_4$$

2. Upon hydrogen desorption, Mg2NiH4 preferentially decomposes to release hydrogen and introduces disordered regions and mechanical stress at the interface, thereby accelerating the decomposition of MgH2.

The "Hydrogen pump" pair Mg_2Ni/Mg_2Ni_4 will play a critical role into the hydrogen desorption/absorption of MgH_2+ $Ni(acac)_2$, as will be discussed later. [4]

2.5 Nickel complex catalyst precursor: Ni(acac)2

Shen *et al.* developed a one-step high-energy ball milling method to generate ultrafine Ni nanoparticles in situ within the MgH_2 matrix using nickel acetylacetonate ($Ni(acac)_2$) as a precursor together with expanded graphite (EG) [4]. The in situ formed Ni nanoparticles significantly enhanced the hydrogen desorption kinetics of MgH_2. By investigating MgH_2 + x wt.% $Ni(acac)_2$ (x = 1, 3, 5, and 10), they found that 3 wt.% $Ni(acac)_2$ provided the best balance between dehydrogenation kinetics and hydrogen storage capacity, retaining approximately 6.9 wt.% H_2 during the second dehydrogenation cycle at 300 °C. Isothermal desorption tests at 275 °C further confirmed that $Ni(acac)_2$ markedly accelerated the dehydrogenation kinetics compared with ball-milled MgH_2 [4].
The addition of expanded graphite (EG) further improved the cycling stability of the Mg/MgH_2 system. The optimized MgH_2–Ni–EG nanocomposite released 7.03 wt.% H_2 within 8.5 minutes at 300 °C after 10 cycles, while reducing the dehydrogenation activation energy by 26.9% compared with pure MgH_2. The composite also demonstrated good hydrogen absorption performance, taking up 2.42 wt.% H_2 within 1 hour at 50 °C. These results demonstrate that the combined use of in situ formed Ni nanoparticles and EG is an effective strategy for significantly enhancing the hydrogen storage performance of MgH_2 [4].

The evolution process of Ni(acac)2 can be described as follows: [4]

$$MgH_2 + Ni(acac)_2 \rightarrow MgO + Ni$$

$$MgH_2 + Ni \rightarrow Mg_2Ni + H_2$$

$$Mg_2Ni + H_2 \rightarrow Mg_2NiH_4$$

This includes the hydrogen pump MgNi/MgNiH4.

2.6 Nickel complex catalyst precursor: Ni(OH)

Chen *et al.* investigated a NiO/Ni(OH)$_2$ heterostructure catalyst to enhance the hydrogen generation performance of Mg through hydrolysis [22]. Among the catalysts tested (NiO/Ni(OH)$_2$-2, NiO, Ni(OH)$_2$, and NiO + Ni(OH)$_2$, all at 2 wt.%), NiO/Ni(OH)$_2$-2 exhibited the best performance, producing 880.2 mL g^{-1} of H_2 within 20 s at 30 °C, corresponding to nearly 100% Mg conversion. Even at 0 °C, the composite generated 820.1 mL g^{-1} of H_2 within 60 s, with a maximum hydrogen generation rate of 4199.9 mL g^{-1} min^{-1}. Ball milling also resulted in a homogeneous distribution of Ni and O within the Mg matrix, providing diffusion pathways that further enhanced hydrolysis kinetics [22].

The superior catalytic performance was attributed to strong interfacial electronic interactions between NiO and Ni(OH)$_2$. Electron transfer from Ni(OH)$_2$ to NiO creates a built-in electric field at the heterostructure interface, accelerating electron transfer and promoting water dissociation. This facilitates hydrogen formation, breaks the water dissociation equilibrium, and significantly accelerates the hydrolysis reaction, enabling nearly complete conversion of Mg to hydrogen [22]

Due to the promising hydrogen generation due to hydrolysis, Ni(OH) can be promising for hydrogen storage properties, following the reasoning of the hypothesis of [17].

2.7Nickel complex catalyst precursor: phtalocyanine [23]

Transition metal phthalocyanines (TMPcs) are highly stable compounds with catalytic, semiconducting, and magnetic properties arising from a phthalocyanine ring coordinated to a transition metal ion. Their electronic properties can be tuned by changing the central metal ion or through electron/hole doping. Despite extensive research on TMPcs, only limited studies have explored their catalytic effects on hydrogen storage materials such as metal hydrides [23].

Zhang et al. [23] were the first to investigate the catalytic effect of nickel phthalocyanine (NiPc) on the dehydrogenation of MgH_2. Their results showed that adding 10 wt.% NiPc significantly reduced the dehydrogenation temperature by more than 90 °C compared with milled pristine MgH_2 and graphene-added MgH_2. The MgH_2 + 10 wt.% NiPc composite exhibited an onset dehydrogenation temperature of 310.4 °C, approximately 99 °C and 94 °C lower than those of milled pristine MgH_2 and MgH_2 + 10 wt.% graphene, respectively. Differential scanning calorimetry and thermogravimetric analyses confirmed that this improvement was due to the catalytic activity of NiPc rather than its decomposition, demonstrating that NiPc effectively promotes MgH_2 decomposition at lower temperatures [23].

Although NiPc exhibits excellent catalytic activity for enhancing the dehydrogenation of MgH_2, its ability to confine MgH_2 particles and prevent their aggregation is relatively weak [23].

2.8 Nickelocene

Zhang *et al.* [24] synthesized nano-LiBH$_4$@Ni composites consisting of ultrafine LiBH$_4$ nanoparticles decorated with ~3 nm Ni clusters using a one-pot solvothermal method. Commercial chemicals, including n-BuLi (2.0 M in cyclohexane, Sigma-Aldrich), (C2 H 5) 3 NBH 3 (97%, Aladdin), Cp2Ni (98%, Aladdin) and n-hexane (97.5%, SuperDry, Acros Organics), were purchased and used as received. During dehydrogenation, B and LiH clusters (5–10 nm) formed while the Ni clusters remained stable. The Ni nanoclusters catalyzed H_2 dissociation into atomic hydrogen and weakened B–B bonds through strong interaction with B clusters, enabling the rehydrogenation of B/LiH back to LiBH$_4$ at temperatures as low as 30 °C under 100 bar H_2. The hydrogenation process proceeds through H_2 dissociation on Ni, formation of borohydride intermediates ($Li_2B_5H_5$, $Li_2B_7H_7$, and $Li_2B_3H_8$), and diffusion of H atoms into the bulk to regenerate LiBH$_4$ [24].

Peng *et al.* [25] enhanced the hydrogen storage performance of MgH_2 by generating highly dispersed

Ni nanoparticles through the *in situ* hydrogenolysis of nickelocene ($NiCp_2$) during ball milling. The MgH_2–16.1 wt.% $NiCp_2$ composite exhibited the lowest hydrogen desorption onset temperature (225 °C), compared with MgH_2–5.62 wt.% Ni (250 °C) and pristine MgH_2 (320 °C). Ball milling under 4 MPa H_2 produced uniformly dispersed ~8 nm Ni nanoparticles within the MgH_2 matrix. During the first dehydrogenation cycle, these Ni nanoparticles reacted with MgH_2 to form finely dispersed Mg_2Ni, which significantly improved reversible hydrogen absorption/desorption kinetics. The highly dispersed Ni-based catalytic phase also provided excellent cycling stability, retaining approximately 96% of its hydrogen storage capacity after 50 cycles [25].

2.9 Ni3PCy3 and the Hindering Effect of Cl

Galey *et al.* [2] investigated the effect of the Ni_3PCy_3 complex on the hydrogen storage performance of MgH_2 by preparing ball-milled composites with varying $NiPCy_3$ contents, PCy_3, and Ni or Fe nanoparticles. Doping MgH_2 with only 5 wt.% $NiPCy_3$ (0.42 wt.% Ni) significantly enhanced hydrogen storage performance, reducing the hydrogen desorption onset temperature to 220 °C and enabling the absorption of 6.2 wt.% H_2 within 5 min at 200 °C under 30 bar H_2. However, this performance deteriorated upon cycling due to the decomposition of the $NiPCy_3$ complex. The addition of Ni and Fe nanoparticles improved cycling stability, allowing hydrogen release below 200 °C, rapid absorption of 4 wt.% H_2 within 5 min at 100 °C, and good reversible hydriding/dehydriding performance. The study concluded that the $NiPCy_3$ complex enhances MgH_2 performance by inhibiting crystal growth, catalyzing hydrogen release, and promoting the homogeneous dispersion of Ni on the Mg/MgH_2 surface [2].

Increasing the $NiPCy_3$ content to 20 wt.% did not further improve dehydrogenation kinetics; instead, hydrogen was released at higher temperatures than with the 5 wt.% $NiPCy_3$ sample. This unexpected result was attributed to the higher chlorine content in the $NiPCy_3$ complex, which acts as a kinetic inhibitor for MgH_2 decomposition and offsets the beneficial catalytic effect of the additional Ni [2].

As of the nickel complexes I will use contains Cl, this kinetic inhibiting effect may have to be considered.

3.Related Works: MgH2 +Ni+ 2D Nanosheets

3.1 Graphene

Schneemann *et al.* [6] reviewed nano-encapsulation strategies for hydrogen storage materials, highlighting graphene, graphene oxide, and poly(methyl methacrylate) (PMMA) as effective encapsulating materials. These coatings allow hydrogen to diffuse through while blocking other gases, protecting metal hydrides from contamination, stabilizing nanoparticles, and improving hydrogen absorption and desorption behaviour [6].

Physisorption materials (e.g., graphene) suffer from weak hydrogen binding and poor ambient-temperature storage, whereas chemisorption materials (e.g., metal hydrides) exhibit strong hydrogen bonding, resulting in slow kinetics and high dehydrogenation temperatures. Combining both approaches into functional composites offers a promising solution, where metal hydrides provide high hydrogen capacity and physisorption materials act as supports to enhance kinetics and reversible hydrogenation. Nano-encapsulation using graphene, reduced graphene oxide, polymers, or MXenes also minimizes dead space compared to scaffolded structures, enabling higher metal hydride loading and greater hydrogen storage capacity [13].

Zhang *et al.* [27] synthesized highly dispersed MgH_2 nanoparticle–graphene nanosheet composites by *in situ* growing MgH_2 nanoparticles on graphene. The graphene confinement effect maintained ~3 nm MgH_2 nanoparticles with high dispersion, leading to outstanding hydrogen storage performance. The composite exhibited an onset dehydrogenation temperature of 255 °C, released 5.1 wt.% H_2 within 20 min at 325 °C, and reabsorbed 5.2 wt.% H_2 within 10 min at 250 °C under 2 MPa H_2. In addition, the activation energy decreased to 118.9 kJ mol^{-1}, approximately 90 kJ mol^{-1} lower than that of commercial MgH_2, demonstrating significantly enhanced hydrogen desorption kinetics due to the graphene confinement effect [27].

Similarly, magnesium nanoparticles coated with nitrogen-doped graphene nanoribbons (GNRs) achieved a hydrogen storage capacity of 7.1 wt.% with

substantially improved absorption and desorption kinetics. The enhanced performance was attributed to the stabilization of surface defects, reduced energy barriers, and improved charge transfer caused by nitrogen doping, which strengthened the interaction between magnesium and hydrogen [13, 26].

Graphene-based materials are promising additives for magnesium-based hydrogen-storage systems because they can accelerate hydrogen release while suppressing oxidation and particle agglomeration without substantially reducing gravimetric hydrogen capacity. [34] For instance, Mg encapsulated within reduced graphene oxide nanolaminates showed improved oxidation resistance and released approximately 5.8 wt% H_2 within 78 min at 300 °C without an observable incubation period. [35] Ball-milled MgH_2–rGO nanocomposites also exhibited a reduction of approximately 30 °C in the nonisothermal hydrogen-release temperature. Their enhanced isothermal desorption behavior was associated with one- and two-dimensional growth of the Mg phase within MgH_2.[36] In a subsequent study, approximately 87% of the stored hydrogen was released within 1 min at 300 °C.[37]This rapid desorption was attributed primarily to the formation of additional interfacial boundaries between carbon edges and MgH_2, which provided efficient pathways for hydrogen diffusion.[38]These findings suggest that Mg–C interactions play an important role in improving hydrogen-release behavior and therefore require further investigation.[37]

3.2 HB

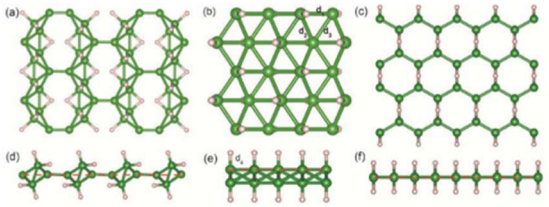


Figure 5 Structure of stabilized hydrogen boride sheets (Green: B, Pink: H) (a) (c) Image viewed from above the seat surface, (d) - (f) Side view of the sheet (a), (d) C2/m, (b), (e) Pbcm, (c), (f) Cmmm. Image is taken from [1]

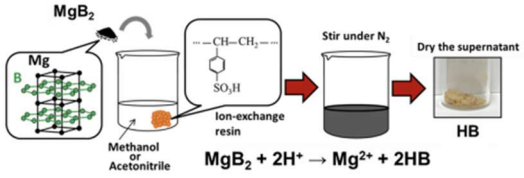


Figure 6 Synthesis process of hydrogen boride sheets. [8]

Hydrogen boride (HB) sheets are a novel two-dimensional boron-based material derived from MgB_2, a layered compound consisting of hexagonal boron sheets separated by Mg cations [8]. Owing to boron's low density, high strength, heat resistance, and high melting point, Kondo *et al.* developed a top-down ion-exchange method in 2017 to synthesize HB sheets by replacing Mg^{2+} ions in MgB_2 with protons using a strongly acidic ion-exchange resin in nonaqueous media under ambient conditions [1,8]. The process produced exfoliated HB nanosheets with a reproducible yield of approximately 42.3%, while the main by-product, $B(OH)_3$, was removed through purification [8].

HB sheets consist of a hexagonal boron network with hydrogen atoms bonded through three-centre two-electron (B–H–B) and two-centre two-electron (B–H) bonds, with an approximate H:B ratio of 1:1 [8,9]. The boron atoms are negatively charged while the hydrogen atoms possess a proton-like positive charge, distinguishing HB from conventional metal borohydrides. The nanosheets are less than 0.5 nm thick and exhibit good chemical stability in water [1,9].

As reported by Kondo et.al [8] in 2017, the use of ionexchange resins for MgB2 enabled them to synthesize hydrogen boride sheets, which are two-dimensional nanosheets composed of hydrogen and boron. The synthesis process is showed in Figure 6. .By reacting MgB2 with a strongly acidic cation exchange resin in acetonitrile or methanol, magnesium ions, and protons are ion-exchanged, and then exfoliated, two-dimensional nanosheets (hydrogen boride sheets) were successfully synthesized. This entire process was conducted under nitrogen at room temperature and ambient pressure with stirring. After 3 days, black precipitates were removed by filtration, and the filtrate was dried on a hot plate at 343 K under nitrogen. HB sheets were then obtained as a yellow powder as the ion exchange reaction product at an average reproducible yield of 42.3%. The main byproduct was B(OH)3 ,

whichwas formed by the hydrolysis reaction of HB sheets with waterabsorbed in the ion-exchange resin . The HB sheets were purified by filtration after cooling the acetonitrile suspension to 255 K . In the methanol suspension, B(OH)3 reacts with methanol to form trimethyl borate, which evaporates under the drying process at 343 K.

As reported by Kondo et al. [8] in 2017, the use of ion exchange resins for MgB_2 enabled them to synthesize hydrogen boride sheets, which are two-dimensional nanosheets composed of hydrogen and boron. The synthesis process is showed in Figure 6. By reacting MgB_2 with a strongly acidic cation exchange resin in acetonitrile or methanol, magnesium ions and protons are ion-exchanged, and then exfoliated, two-dimensional nanosheets (hydrogen boride sheets) were successfully synthesized. This entire process was conducted under nitrogen at room temperature and ambient pressure with stirring. After 3 days, black precipitates were removed by filtration, and the filtrate was dried on a hot plate at 343 K under nitrogen. HB sheets were then obtained as a yellow powder as the ion exchange reaction product at an average reproducible yield of 42.3%. The main byproduct was $B(OH)_3$, which was formed by the hydrolysis reaction of HB sheets with water absorbed in the ion-exchange resin. The HB sheets were purified by filtration after cooling the acetonitrile suspension to 255 K. In the methanol suspension, $B(OH)_3$ reacts with methanol to form trimethyl borate, which evaporates under the drying process at 343 K.

HB sheets have demonstrated a range of attractive properties, including solid-acid catalysis, gas sensing, semimetallic behavior, light-induced hydrogen release, and CO_2 adsorption and conversion [9]. They also serve as effective supports and reducing agents for metal nanoparticles, enabling the reduction and immobilization of metals such as Pd, Pt, Ni, Co, Zn, and Cu [1]. Furthermore, HB-assisted synthesis has produced highly dispersed Pt nanoparticle electrocatalysts with superior catalytic activity and stability compared with commercial fuel-cell catalysts [1,10].

In this research project, HB or graphene will be investigated as a 2D support material for Ni catalysts and MgH_2 to improve hydrogen storage performance [1].

In research, either HB or graphene will be chosen as the 2D-material to support the Ni catalyst and MgH2.

3.3 Ni/NB

In her Master's thesis, Noguchi created MgH2 -Ni/HB composites and investigate their hydrogen storage-release mechanism [1]. Ni nanoclusters were successfully dispersed on HB sheets.

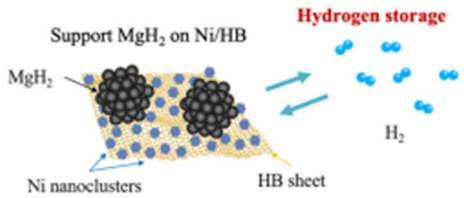


Figure 7 Schematic image of the conception of Noguchi's work in master's thesis [1]

As a future reference, Noguchi [1] synthesised Ni/HB using the following process with Powder HB 95 mg / acetonitrile 80 mL. (Figure 8) $Ni(CH_3COCHCOCH_3)_2$ 206.6 mg, 102.7 mg, 10.7 mg / acetonitrile 80 mL

1. HB acetonitrile dispersion and Ni complex acetonitrile solution were prepared by adding each to acetonitrile.
2. The synthesis was started by adding HB acetonitrile dispersion to Ni complex acetonitrile solution.
3. In order to degas the air in acetonitrile, "pull vacuum for one minute → introduce argon gas to normal pressure" is repeated for three sets.
4. Stirred at 310 rpm for about 24 hours under an Ar atmosphere. The mixture was covered with a double layer of black plastic bags to shield it from light.
5. After about 24 hours, the stirrer was removed and concentrated and dried while pulling a vacuum. During the concentration, the mixture was immersed in an oil bath at 30-45°C to avoid frost formation. During drying, the temperature was increased by 5 °C in increments to 70 °C.

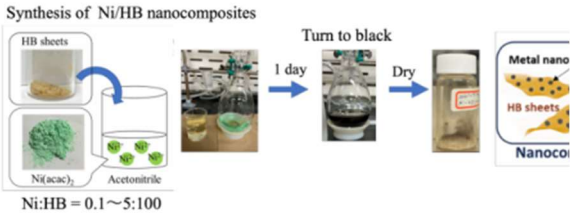


Figure 8 Schematic image of synthesis method of Ni/HB sheets. Image is from [1]

MgH_2–Ni/HB composites exhibit improved hydrogen desorption performance compared with ball-milled MgH_2, owing to the combined effects of Ni nanoclusters and hydrogen boride (HB) sheets. The $MgH_2(Ni_{0.1}/HB_1)$@Ni 2.0 wt.% composite showed the lowest hydrogen release temperature of 550 K, approximately 50 K lower than that of ball-milled commercial MgH_2 (600 K). Composites containing $Ni_{0.005}$/HB and $Ni_{0.05}$/HB were also evaluated, but $Ni_{0.1}$/HB provided the greatest reduction in desorption temperature while maintaining the lowest Ni loading (2 wt.%) [1].

The enhanced performance was attributed to Ni nanoclusters acting as hydrogen recombination catalysts, while the MgH_2–Ni interface facilitated hydrogen transfer. The stepwise decrease in hydrogen desorption temperature with increasing Ni content suggests that intimate contact between Ni and MgH_2 promotes efficient hydrogen hopping and accelerates dehydrogenation. In addition, HB sheets help suppress MgH_2 aggregation, further contributing to the lower hydrogen release temperature. Differential scanning calorimetry (DSC) measurements also confirmed the reversible hydrogen storage behavior of the composites, with hydrogen desorption occurring at 350 °C and hydrogen absorption at 50 °C [1].

Previous studies have primarily investigated individual Ni-based catalyst precursors for improving the hydrogen storage properties of MgH_2. In contrast, this research proposes a systematic comparison of multiple nickel complex precursors by preparing MgH_2–Ni composites with and without HB or graphene under identical ball-milling conditions. The hydrogen storage capacity, dehydrogenation kinetics, and cycling performance of the MgH_2–Ni and MgH_2–Ni/HB (or graphene) composites will be directly compared to identify the most effective catalyst system.

Currently, five nickel complex candidates have been selected. NiPc has been characterized using FTIR, while $Ni(acac)_2$ has been investigated as the first catalyst precursor for MgH_2 dehydrogenation. The nickel complexes were selected based on (1) the nearest atom, ion, or ligand directly coordinated to Ni, which may influence catalyst–MgH_2 interactions and hydrogen storage behavior, (2) molecular size (molecular weight), and (3) commercial availability (such as sold by Fujifilm or Sigma Aldrich).

For quicker H release from MgH_2, Mg should be negative, and for quicker H absorption, Mg should be positive. [15] Therefore, the electronegativity of the elements in the nickel complexes play an important factor because they determine the charge of Mg when binding after MgH_2+Nickel complex are ball milled into a mixture.

4.Nickel Complexes Candidates

I.Nickel(II) phthalocyanine(($C_{32}H_{16}N_8$)Ni), min. 94%

The available sample we obtain has a mass of 5 g. The molecular structure of the complex can be seen in Figure 2 above.

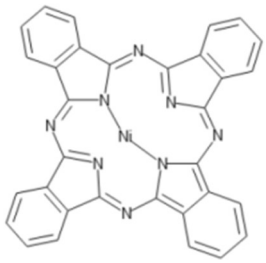


Figure 9 Structure of Nickel(II) phthalocyanine

The ($C_{32}H_{16}N_8$)Ni sample candidate complex has characteristics:

1) The element closest to Ni is N(Nitrogen)
2) The molecular weight is 571.23 amu

II.Nickel acetylacetonate ($Ni(acac)_2$)

The molecular structure of $Ni(acac)_2$ can be seen in Figure 5.

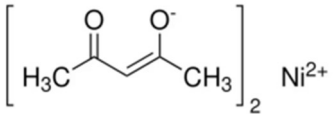


Figure 10 Structure of Ni(acac)2

Nickel (II) acetylacetonate ($Ni(C_5H_7O_2)_2$) has the following characteristics:

1. The compound closest to Ni is acetylacetone, composed of CH3 and O
2. The molecular weight is 256.91 amu

III. $NiCl_2 \cdot 6H_2O$

The molecular structure of Nickel(II) Chloride Hexahydrate is in Figure 11.

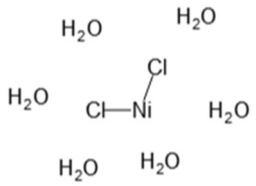


Figure 11 Structure of $NiCl_2 \cdot 6(H_2O)$

$NiCl_2 \cdot 6H_2O$ has the following characteristics:

1. The element closest to Ni is Cl
2. The molecular weight is 237.69 amu.

IV. Ni(OH)

The molecular structure of Nickel(II) Chloride Hexahydrate is in Figure 12.

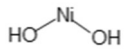


Figure 12 Structure of Ni(OH)

Ni(OH) has the following characteristics:

1. The compound closest to Ni is OH
2. The molecular weight is 237.69 amu.

V. **Nickelocene/CpNi ($(C_5H_5)_2Ni$)**

The molecular structure of CpNi is in Figure 13.

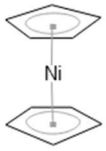


Figure 13 Structure of CpNi

The molecule the following characteristics:

1. Ni is “sandwhiched” between two parallel cyclopentadienyl ($C_5H_5^-$) rings. The carbon atoms of the two cyclopentadienyl rings are the closest to nickel

2. The molecular weight is 188.90 amu.

5.Methodology

The proposed research project is designed to comprehensively investigate hydrogen storage capacity, hydrogen absorption-desorption behaviour, and kinetics of MgH2 -Ni composite from nickel complex salts formed through ball milling. This study also aims to explore the fundamental mechanisms by which ball milling, a process of mechanical alloying, influences the microstructural, morphological, and hydrogen storage characteristics of the MgH2 -Ni composite. The catalyst activity of Ni and the role of MgH2 -Ni interface will also be investigated. A 2D-nanosheet (HB or graphene) will be added later to form MgH2 -Ni-HB/graphene composites with each different nickel complex salt. Then the hydrogen storage capacity, hydrogen absorption-desorption behaviour, kinetics, Ni catalyst activity, role of MgH2-Ni, effect of addition of HB/graphene, and reaction pathways of the MgH2 -Ni-HB/graphene composite will be compared with MgH2 -Ni-HB/graphene.

5.1 Sample Preparation

Nickel complex salts, MgH_2, and HB or graphene (2D material to be decided later through discussion) will be used as raw materials. First, MgH_2 and each of the nickel complex salts will be mixed and ball-milled (a mechano-chemical process) to form MgH_2–Ni composites. We use a planetary ball mill manufactured by Retsch. The variables of interest include milling time (to assess the impact of prolonged mechanical energy input), ball-to-powder ratio (to determine the optimal mass ratio for efficient milling), and milling speed.

The ball-to-powder mass ratio is 48:1.

We will use two methods to synthesise MgH_2–Ni composites in our experiment and then compare the results using characterisation techniques. The two methods are as follows:

1. One-step ball milling method: Ball mill MgH_2 + NiX
2. Two-step ball milling method: Ball mill MgH_2 → add NiX → ball mill MgH_2 + NiX

Later, HB or graphene may also be introduced into MgH_2 together with each of the nickel complex salts and then ball-milled to form MgH_2–Ni–HB/graphene composites.

5.2 Characterisation

To ensure a comprehensive analysis of the milled samples, an array of analytical techniques will be employed for this research (Table 1).

Characterisation techniques

Method	Function/Overview
XRD	Crystallinity
IR	Bonding
SEM	Morphology
TG-DTA&MASS	H_2 release temperature,activation energy
PCT	Advanced H_2 capacity

The purpose of each method is below. A detailed explanation of each principle will be attached to future PBL report.

Characterization Methods

X-ray Diffraction (XRD). XRD is a non-destructive technique used to determine the crystal structure, phase composition, and structural changes of hydrogen storage materials. It is widely applied to identify hydride phases, quantify phase abundances, and monitor phase transformations during hydrogen absorption and desorption, providing essential information on the bulk properties of metal hydrides such as MgH_2 [11].

Infrared Spectroscopy (IR) IR spectroscopy identifies the chemical structure and bonding of materials by measuring infrared absorption associated with molecular vibrations. Because each compound has a unique vibrational spectrum, IR is particularly useful for characterizing nickel complexes and confirming their molecular structures. Infrared absorption occurs only when molecular vibrations produce a change in dipole moment, making IR effective for identifying functional groups and chemical bonding environments [1].

Molecules absorb IR radiation when its energy matches the energy required to excite molecular vibrations. Since different chemical bonds have different bond strengths, each functional group absorbs IR radiation at characteristic frequencies, allowing them to be identified from the resulting spectrum.

The vibrational IR region commonly used in chemistry ranges from 4000 to 400 cm^{-1}, expressed as wavenumbers, which are directly proportional to the energy of the radiation. Covalently bonded atoms continuously vibrate, and these vibrational transitions occur when the molecule absorbs IR radiation.

For a vibration to be IR-active, the vibrating bond must be polar and its vibration must produce a change in the bond dipole moment. Generally, the greater the bond polarity, the stronger the IR absorption. IR spectra are displayed with absorbance (or transmittance) on the vertical axis and wavenumber (cm^{-1}) on the horizontal axis. The observed absorption peaks are assigned to specific functional groups by comparing them with standard correlation tables or reference spectra.

Scanning Electron Microscopy (SEM). SEM provides high-resolution images of the surface morphology and microstructure of hydrogen storage materials. It is used to examine particle size, surface roughness, porosity, agglomeration, and structural defects, all of which influence hydrogen diffusion, adsorption, and long-term cycling stability. SEM is therefore valuable for correlating microstructural changes with hydrogen storage performance [11].

Thermogravimetric–Differential Thermal Analysis coupled with Mass Spectrometry (TG–DTA–MS). TG–DTA simultaneously measures weight changes (TG) and thermal events (DTA) during heating, allowing evaluation of hydrogen desorption temperatures, reaction enthalpies, and whether reactions are endothermic or exothermic [1,27,28]. Thermogravimetry is performed using a thermobalance. As the temperature rises, convection and buoyancy forces around the balance cause an apparent weight change. A mass loss is associated with an endothermic reaction whereas mass gain is associated with an exothermic reaction [27] In the differential type,

buoyancy and convection forces are applied to both the sample holder and the reference holder, and these effects are canceled out .Differential thermal analysis (DTA) detects the thermal change in a sample due to temperature change as a temperature difference from a reference material, and the DTA signal provides information on the transition temperature and reaction temperature of the sample, and whether the transition or reaction is endothermic or exothermic. [1]] Usually in DTA, if the transition or reaction is endothermic, a downward peak is obtained, whereas if it is exothermic, an upward peak is obtained [28], and the area under the endotherm/exotherm is related to the enthalpy in TG-DTA.

Coupling TG–DTA with quadrupole mass spectrometry (Q-MS) enables identification of the gases evolved during heating by measuring their mass-to-charge ratio (m/z). In the TG-DTA-MS, the evolved gases from the interface are injected towards the mass chamber prior to ionization in the ionization room of the QMS, then the gas ions pass through between the quadrupoles, detected by the secondary electron multiplier and outputted as an ion current value. [27]

The combined TG–DTA–MS technique therefore provides comprehensive information on hydrogen desorption behaviour, gas evolution, and reaction mechanisms [12,27].

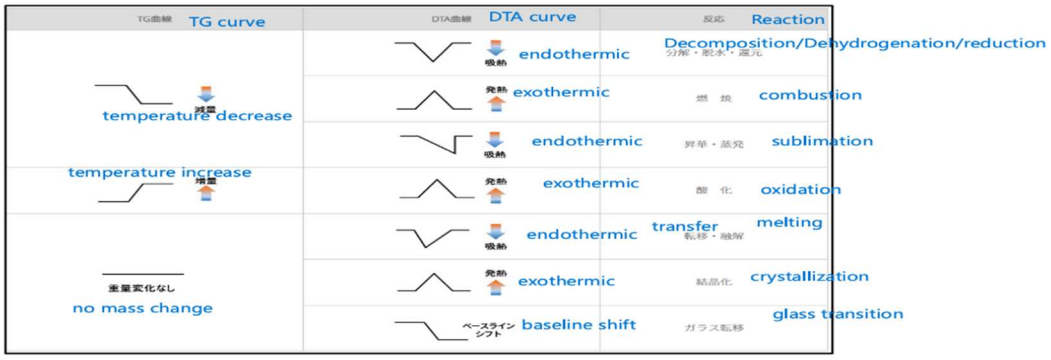


Figure 14 Reactions associated with TG-DTA curve orientation. From [28]

Sample example of TG curve orientation, DTA curve orientation and types of reaction is in Figure 14. From [28], translated to English.

The evolved gases from the sample room of the TG DTA at atmospheric pressure will be injected towards the MS chamber under high vacuum pressure via the interface. The interface is a structure where it essentially creates a difference in pressure between the TG-DTA and the MS chamber requiring various innovative devices. The interface is classified into two types, the capillary type and the skimmer type.

A figure of the capillary type is below: Image in Figure 15. Image is from [27]

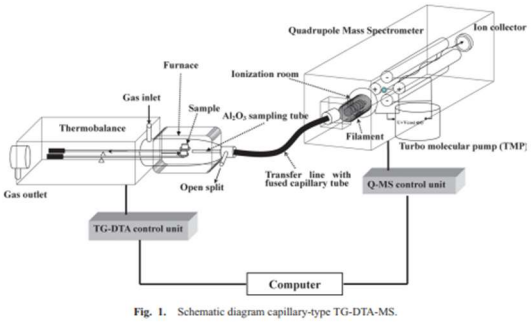


Fig. 1. Schematic diagram capillary-type TG-DTA-MS.

Figure 15 Diagram of capillary TG-DTA-MS

- When a mass loss is observed on the TG curve of the TG-DTA, then gas components have volatized from the sample.

At this time, the gases evolving instantly within a short period of time has a good measurement sensitivity than gases evolving slowly at a wide temperature range even if the same amount of gases have evolved because the MS detects the amount of evolved gases per unit time. However, a higher heating rate leads to a better sensitivity in the measurement of the amount of evolved gases but leads to poor peak resolution. In case where minute amount of gases evolve from the sample, increasing the heating rate will enhance the sensitivity of he

measurement, expecting some amount of gases while a slow heating rate will increase the peak resolution. Since the MS data obtained from the evolved gases in the sample shows the gas evolution rate, the TIC (total ion current) of the MS and the derivative TG (DTG) curve have a similar relationship. [27]

Pressure–Composition–Temperature (PCT) measurements. PCT analysis evaluates the hydrogen storage capacity, equilibrium pressure, and thermodynamic behavior of hydrogen storage materials under isothermal conditions [11]. Combined with the van't Hoff equation, PCT measurements enable the determination of reaction enthalpy and entropy, providing insight into the thermodynamics of hydrogen absorption and desorption. Because PCT is more time-consuming and advanced than the other characterization techniques, it will be used primarily for the most promising MgH_2–Ni composites identified during the initial screening [11].

The Vant Hoff equation

$lnK = -\Delta H/RT + \Delta S/R$ (1)

where K is the equilibrium constant, H is the standard reaction enthalpy (kJ/mol), S is the standard reaction entropy (J/mol·K), R is the gas constant (J/mol·K), and T is the temperature (K), reveals the relationship between the chemical equilibrium constant and temperature, forming the theoretical foundation for studying the effect of temperature on equilibrium in reaction thermodynamics.

In our initial step to optimise the ball-milling conditions for preparing MgH–Ni composites using Ni(acac)2, we will first do characteristations XRD and IR before and after ball milling. After ball milling, we will also do dehydrogenation examination using TG-DTA and TPD, and post-dehydrogenation characterisations XRD and IR.

Our current initial workflow is below:

Before ball milling	XRD	IR
	Ball mill	
	TG-DTA, TPD	

6.Experiment

6.1 Sample preparation

Commercial MgH_2, commercial nickel acetylacetonate, ($Ni(acac)_2$,)$_2$), and commercial nickel phthalocyanine (NiPc) from Strem Chemicals were obtained and used without further purification. All procedures were carried out in an argon-filled glovebox to prevent contamination of the samples. First, MgH_2 was ball-milled with seven balls at 400 rpm, using a ball-to-sample mass ratio of 48:1, for 4 hours (44 cycles) in a Fritsch planetary ball mill. To minimize temperature rise during prolonged milling, each cycle consisted of 5 minutes of rotation followed by a 30-second rest period. All milling and handling of the powders were performed inside an argon-filled glovebox.

Then, we synthesised MgH_2 + 2 wt.% $Ni(acac)_2$ using two different ball-milling methods, referred to as the "Intensive" and "Gentle" methods, for comparison.

1. Intensive method:

We doped 2 wt.% $Ni(acac)_2$ into commercial MgH_2 using a Fritsch planetary ball mill at 400 rpm for 4 hours (44 cycles) under an argon atmosphere with four balls. The ball-to-sample mass ratio was 48:1, and each cycle consisted of 5 minutes of rotation followed by a 30-second rest period.

2. Gentle method:

This method employs a two-step ball-milling process using gentler milling parameters to preserve the structure of $Ni(acac)_2$. First, MgH_2 was ball-milled using the same parameters as the previous samples: 400 rpm, a ball-to-sample mass ratio of 48:1, for 4 hours (44 cycles) in a Fritsch planetary ball mill. Each cycle consisted of 5 minutes of rotation followed by a 30-second rest period. Next, 2 wt.% $Ni(acac)_2$ was added to the pre-milled MgH_2 and ball-milled in a Fritsch planetary ball mill at 400 rpm for 2 hours (24 cycles) under an argon atmosphere using four balls. The ball-to-sample mass ratio was 48:1, and each cycle consisted of 5 minutes of rotation followed by a 30-second rest period.

The comparison of each sample's ball milling parameters are summarised in Table 2.

Table 1 Ball milling Conditions

Parameters	MgH_2	Intensive MgH_2+2 wt.% $Ni(acac)_2$	Gentle MgH_2+2 wt.% $Ni(acac)_2$
Speed	400	400	1st step (MgH_2): 400, 2nd step: 250
Balls	7	4	Both: 4
Time	300 sec	300 sec	300 sec
Pause	30 sec	30 sec	30 sec
Cycle	44	44	1st: 44, 2nd: 24
Reverse	On	On	On

6.2 Sample Characterisation

FTIR: Fourier Transform Infrared Spectroscopy

Upon obtainment of NiPc sample, the FTIR of was done inside a glovebox using Bruker Japan FT-IR ALPHA II to characterize bonding of NiPc. A tiny amount of powder NiPc was put on the centre of the IR spectrometer which is inside the glovebox. The IR transmittance then could be detected.

We started the measurement by operating OPUS software in lab PC connected to the FTIR machine. The IR spectra then appeared on screen.

Background subtraction was done on the IR spectra. We also processed the IR spectra using Igor software.

Thermogravimetric-Differential Thermal Analysis-Quadrapole Mass Spectroscopy (TG-DTA-QMS) and Temperature Programmed Desorption (TPD) for Hydrogen Desoprtion Measurement

Thermogravimetric and differential thermal analysis (TG-DTA) was conducted in conjunction with hydrogen temperature-programmed desorption (H_2-TPD) using a STA-2500 Regulus analyzer under argon flow. The exhaust gas was also sampled through a capillary connection and then analyzed for hydrogenation via a quadrupole mass spectrometer (MKS, Microvision 2) in an ultrahigh-vacuum chamber. Using EasyView software connected to the QMS, the detected mass numbers were set to 15 channels for consistency and equal interval: mass numbers 2, 3, 14, 15, 16, 17, 18, 20, 28, 30, 32, 34, 40, 46, 48.

The first sample, ball milled MgH_2 (7 balls) with the mass of 10.9 mg was loaded into an alumina cup for thermal analysis. The second sample, intensively ball milled MgH_2-$Ni(acac)_2$, also loaded into an alumina cup, had a mass of 9.7 mg. The third sample, gently ball milled MgH_2-$Ni(acac)_2$, loaded into an alumina cup, had a mass of 8.8 mg. Hydrogen injection for hydrogen calibration was conducted at room temperature to the STA-2500 Regulus analyzer. For the MgH_2 sample, the hydrogen injection was conducted at 0.01 mL, 0.05 mL, 0.1 mL, 0.2 mL, and 0.5 mL. Meanwhile, for both intensively and gently ball milled samples, hydrogen injection was conducted at 0.01 mL, 0.05 mL, 0.1 mL, and 0.5 mL using a syringe at room temperature. All three samples were heated from 25 °C to 600 °C, with a DTA heating speed of 10 K/min.

Raw TG-DTA data was recorded in NETZSCH data, and TG-DTA data was processed first in an Excel calculation file into TPD (Temperature Programmed Desorption).

X-Ray Diffraction (XRD)

X-ray diffraction (XRD) measurements were conducted at room temperature (~ 25 °C) using a benchtop diffractometer (Rigaku MiniFlex) with Cu Kα radiation at a voltage of 40 kV and current of 15 mA was used to identify the phase of the samples. X-rays were generated using the line focus principle. Samples are placed inside a Rigaku atmospheric separator, and a Kapton film capsule was used to prevent atmospheric exposure.

The XRD data was captured in a 2θ range of $5° \sim 90°$ with a step of $0.0100°$ inside a Rigaku general purpose atmosphere separator. The process of transferring commercial MgH_2, ball milled MgH_2 (4 balls) before TG-DTA, gently ball milled MgH_2 + 2 wt.% $Ni(acac)_2$ before TG-DTA, intensively ball milled MgH_2 + 2 wt.% $Ni(acac)_2$ before TG-DTA, and intensively ball milled MgH_2 + 2 wt.% $Ni(acac)_2$ after TG-DTA into the separator was performed inside an argon glovebox. However, due to the scarcity of the sample which makes the operation more difficult, the process of transferring ball milled MgH_2 (7 balls) after TG-DTA and gently ball milled MgH_2 + 2 wt.% $Ni(acac)_2$ after TG-DTA into the separator was performed exposed to air outside of the glovebox.

For all samples, the measurement condition was:

Start angle	End angle	Step	Speed °/min
5°	90°	0.0100	5.0°/min

The XRD data was scanned in situ using SMARTLab software.

The average crystallite size (D) was estimated using the Scherrer equation:

$D = k\lambda / \beta\cos\theta$ (2)

where k is a dimensionless shape factor (typically 0.89–0.9 for spherical crystals with cubic unit cells), λ is the X-ray wavelength, β is the full width at half maximum (FWHM) in radians (on the 2θ scale), and θ is the Bragg angle.

Current experimental status of our initial experimental plan is in Table 2.

Table 2 Experimental plan checklist of MgH_2 and doping of $Ni(acac)_2$

Ball milled sample	XRD	IR	TG-DTA	XRD	IR
Ball milled MgH_2(7 balls)	✓		✓	✓	
Intensively ball milled MgH_2+ 2 wt.% $Ni(acac)_2$	✓		✓	✓	
1 st step of gentle ball milling of MgH_2+ 2 wt.% $Ni(acac)_2$: Ball milled MgH_2(4 balls)	✓				
2nd step of gently ball milled MgH_2+ 2 wt.% $Ni(acac)_2$	✓		✓	✓	

7 Results and Discussion

7.1 IR Spectra

The IR spectra of NiPc obtained from experiment is in Figure 16.

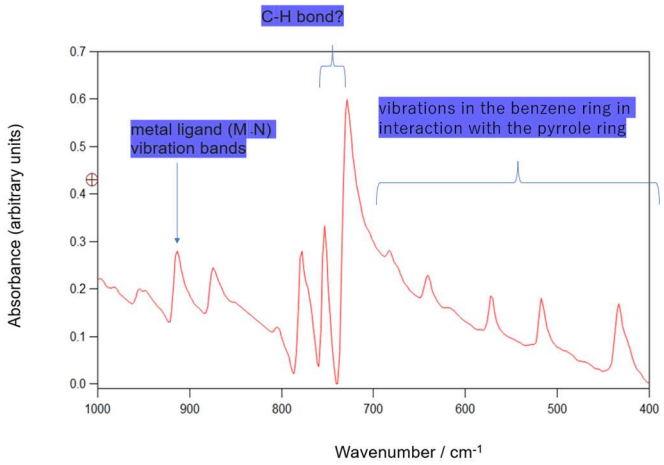


Figure 16 IR spectra of NiPc obtained from experiment

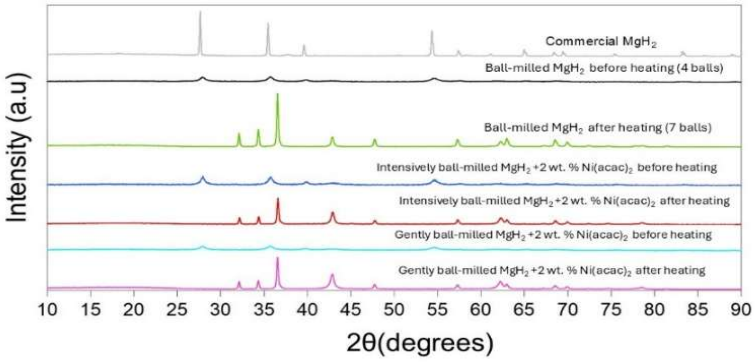
From reference[29], the peaks in the 700–400 cm^{-1} interval originate most probably from vibrations in the benzene ring in interaction with the pyrrole ring. The peak at 753.17 cm-1 may correspond to [29]'s non-planar vibrations (out-of-plane bending) of the C-H bonds at 755 cm−1 and a band at 779 cm−1. Thus, this

peak may correspond to vibration of C-H bond. The peak at 913 cm^{-1} is consistent with metal ligand(M-N) ligand vibration bands observed at frequencies in the range 883-919 cm^{1} for the FePc, CoPc, NiPc, CuPc, ZnPc, PdPc and PtPc.[29]

7.2 XRD

The XRD peaks of all measured samples are in Figure 15. The graph was cut from 2θ=10° since there is a background peak from the kapton film of the atmospheric separator in 2θ=6° (Figure 18)

Figure 17 XRD pattern of samples done so far



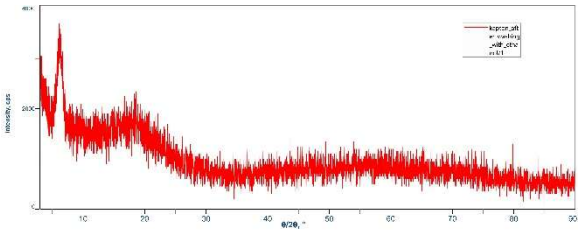


Figure 18 Kapton background peak at 2θ =6°

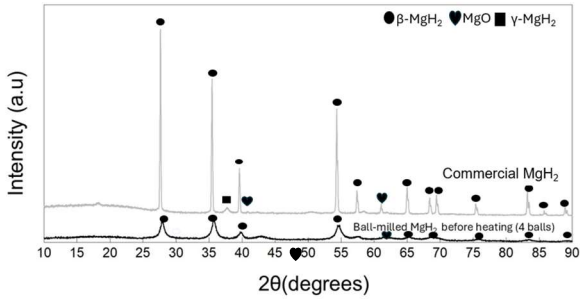


Figure 19 Commercial MgH_2 (gray line) and ball milled MgH_2 (4 balls) (black line)

A comparison of the XRD of commercial MgH_2 and ball milled MgH_2 (4 balls) before heating is in Figure 17. The XRD results from [23] are used as a reference. Ball milled MgH_2 shows broader peaks, which may indicate smaller crystalline size and/or lattice defects in the crystalline structure, or that the sample might be amorphous in nature, a solid lacking perfect crystallinity. [30, 31]. This is reasonable since ball milling decreases crystalline size and introduces defects into the structure. Meanwhile, commercial MgH_2 shows a thinner peak which corresponds to bigger crystal size. [31] The crystalline size and structure of each sample will be further verified and investigated with Scherrer Equation.

Ball milled MgH_2 + 2wt.% $Ni(acac)_2$

Figures 20-23 show XRD results of ball milled MgH_2 + 2wt.% $Ni(acac)_2$ before and after TG-DTA(heating). Figures 1 and 8 from [4] was used as a reference for the peak assignment of the samples. (Figure 24)

For all samples, peaks related to $Ni(acac)_2$ cannot be found. According to [4], this indicates the chemical reaction between MgH_2 and $Ni(acac)_2$ or the decomposition of $Ni(acac)_2$ during the ball milling process.

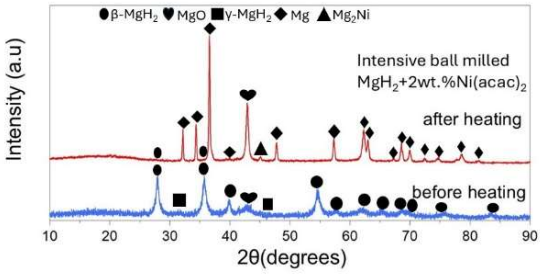


Figure 20 Intensively ball milled sample before and after TG-DTA

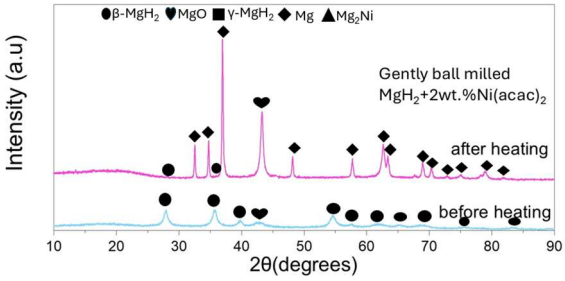


Figure 21 Gently ball milled sample before and after TG-DTA

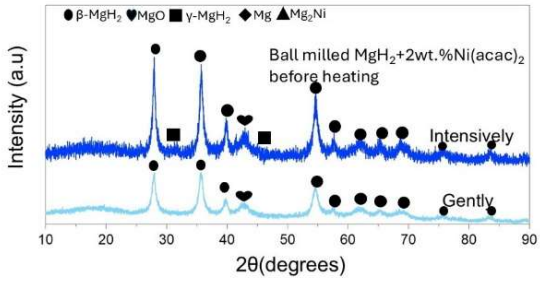
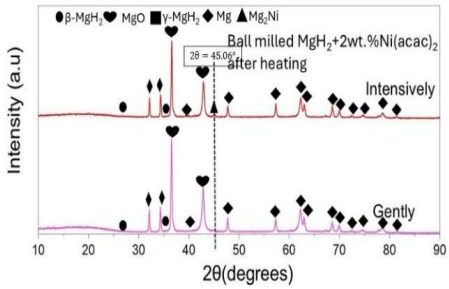


Figure 22 Intensively vs gently ball milled samples before TG-DTA



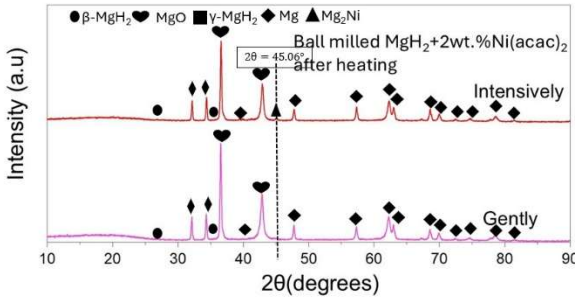


Figure 23 Intensively vs gently ball milled samples after the TG-DTA. At $2\theta = 45.06°$, there is Mg_2Ni diffraction peak in the intensively ball milled sample. However, the peak is absent in the gently ball milled sample.

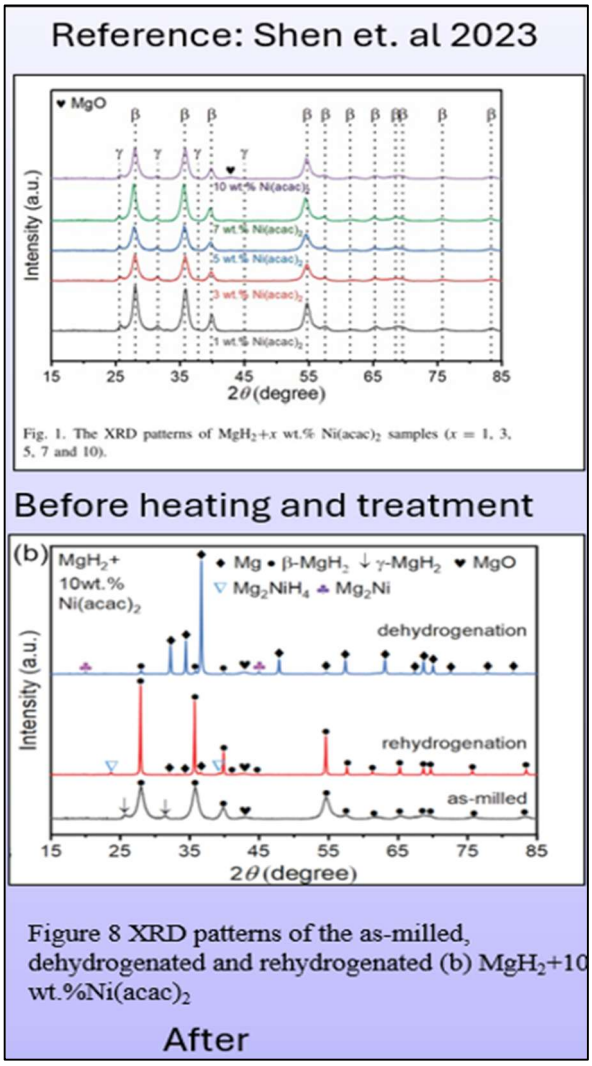


Figure 24 Figures 1 and 8 from [4] used for XRD peak assignment reference

The diffraction peaks of β-MgH_2, γ-MgH_2, and trace amount MgO phases can be found in the intensively ball milled MgH_2–Ni(acac)$_2$ sample before TG-DTA. The intensively ball milled sample and gentle ball milled sample before TG-DTA (heating) show almost identical peaks with the absence of metastable γ-MgH_2 in the gently ball milled sample in Figure 22. According to Reference [4], the formation of the MgO phase may be due to the reaction between MgH_2 and Ni(acac)2 during ball milling.

One interesting finding is the difference of the peaks of the intensively and gently ball milled samples after dehydrogenation (Figure 23). The dehydrogenated gently ball-milled sample and intensively ball-milled sample both have Mg, small amount of MgO and no diffraction peaks of Ni. However, dehydrogenated Ni-

based compounds Mg_2Ni exists in the intensively ball milled sample but not in the gently ball milled sample. The gently ball milled sample may have no Mg_2Ni due to the small content of the Ni phase or the small size in situ formed Ni particles from $Ni(acac)_2$, which might result in the corresponding diffraction peaks being too weak to detect. [4]

With no Mg_2Ni detected in the dehydrogenated gently ball milled sample, the sample lacks the $MgNiMg_2Ni$ "hydrogen pump" which helps promoting better hydrogen desorption performance, as mentioned in Section 2.4 and 2.5 [4, 20, 21]

The crystalline sizes of intensive ball milled and gently ball milled MgH_2 and MgH_2-$Ni(acac)_2$ are calculated by Scherrer's equation. We use X-ray wavelength λ=154 nm due to Cu Kα radiation and a constant k=0.89. The crystalline sizes are graphed against 2θ in Figure 25 for comparison. Ball milled MgH_2(4 balls) intensive ball milled sample is graphed in blue line, while the gently ball milled sample is graphed in light purple line.

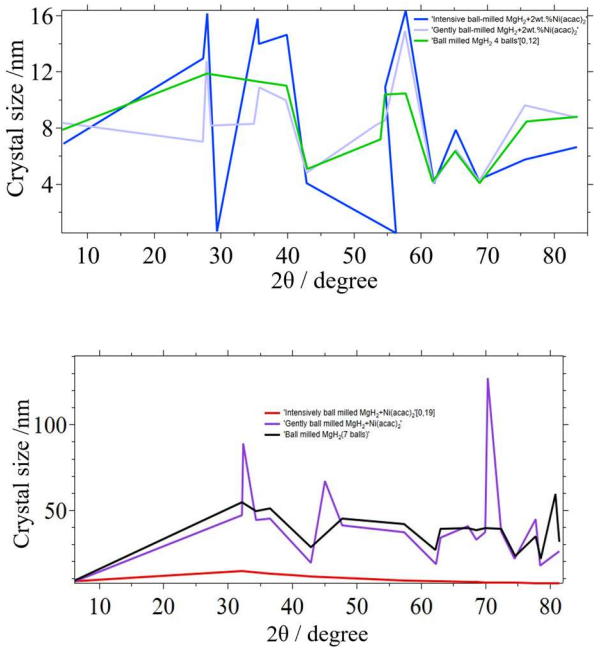


Figure 25 Crystal size (nm) of intensively ball milled and gently ball milled MgH_2+2 wt.% $Ni(acac)_2$. Top figure is before TG

This suggests that the gently ball-milled sample shows smaller crystallite sizes than those of the intensively ball-milled for most corresponding major diffraction peaks, although the trend is not consistent for every angle. The smaller may be because of the 2 step ball milling process.

We also calculated the crystalline size for dehydrogenated samples (samples undergoing TG-

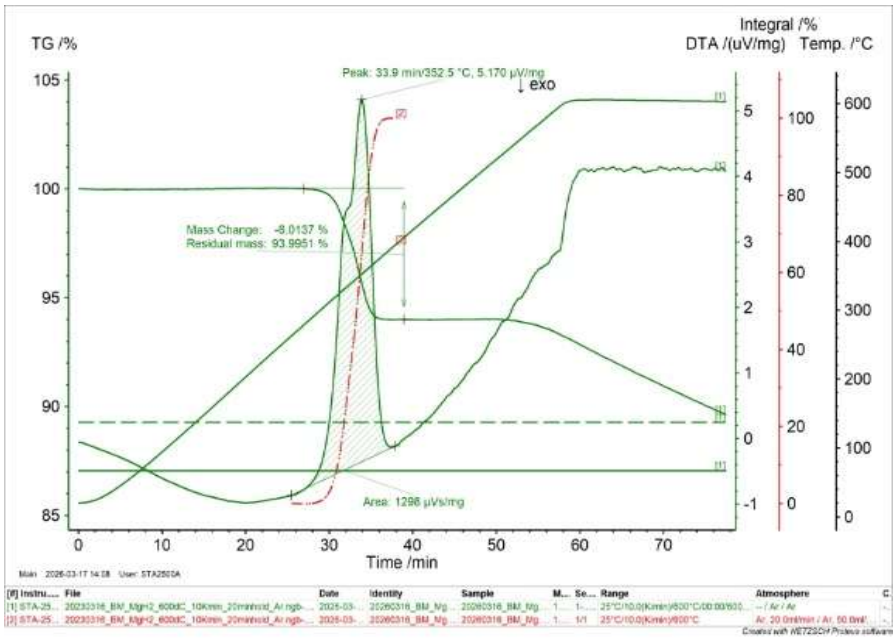


Figure 26 Raw TG-DTA Result of ball milled MgH_2 (7 balls)

DTA). (Figure 25). Ball milled MgH_2(7 balls) (black line), intensively ball milled MgH_2+Ni(acac)$_2$ (red line) and gently ball milled MgH_2+Ni(acac)$_2$.

After TG–DTA, the gently ball-milled composite showed an increase in apparent coherent-domain size,

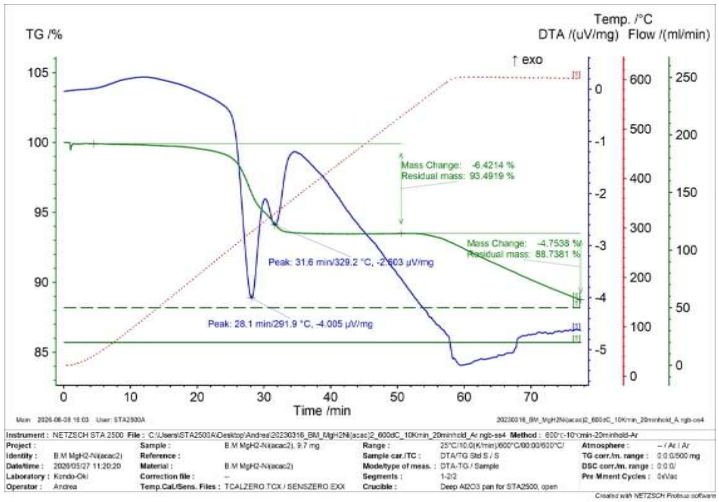


Figure 27 TG-DTA result of intensively ball milled MgH_2+Ni(acac)$_2$

which is attributable to thermal recovery, recrystallization and grain growth during dehydrogenation. In contrast, the intensively ball-milled composite retained broad diffraction peaks and consequently exhibited smaller Scherrer sizes. This behaviour may originate from its higher defect density, residual microstrain and the formation of finely divided Mg-based domains resulting from the intimate dispersion of the Ni-containing additive. However, because the Scherrer method cannot distinguish size broadening from strain and phase-overlap effects, the calculated values should be regarded as apparent coherent-domain sizes rather than direct particle sizes.

7.3TG-DTA and TPD

The raw results of TG-DTA of ball milled MgH_2 (7 balls), intensively ball milled and gently ball milled MgH_2+ Ni(acac)$_2$ are Figures 26,27, and 28 respectively. All 3 samples show weight loss and the opposite direction to exotherm, indicating that all 3 undergo an endothermic reaction during dehydrogenation. Ball milled MgH_2 almost shows one continuous large peak with only a slight off-shoulder peak as its neighbour that goes undetected by the NETZCH software. However, the intensive MgH_2+Ni(acac)$_2$ samples indicate multiple peak and multiple TG weight loss, indicating a multi-step reaction during dehydrogenation.

Based on the TG weight loss calculation of the main peak from the raw data:
Intensively ball milled sample (6.4214% +4.748%=11.594%) >ball milled sample (6.0137%)> gently ball milled sample (5.277%). This may indicate that the intensively ball milled sample releases the highest quantity of H_2 desorption. However, further data processing is needed to isolate only hydrogen components (m/z=2).

The TPD-MS hydrogen release profile (hydrogen mass intensity against temperature) graph of ball milled MgH_2 (7 balls) is in Figure 29.

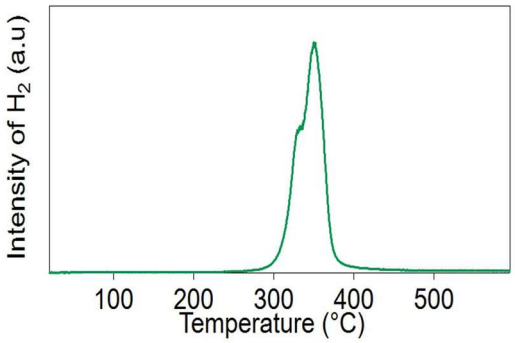


Figure 30 Hydrogen release profile of ball-milled MgH_2(7 balls)

The TPD-MS hydrogen release profile (hydrogen mass Intensity m/z=2 against temperature) graph of ball milled MgH_2 (7 balls) is in Figure 29. The onset temperature is 310°C and the peak dehydrogenation temperature is at 338°C.

Next, the TPD-MS hydrogen release profiles of ball milled MgH_2-$Ni(acac)_2$ are put side by side for comparison in Figure 27. However, for determining the onset and peak dehydrogenation temperatures quantitatively, peak deconvolution may be required.

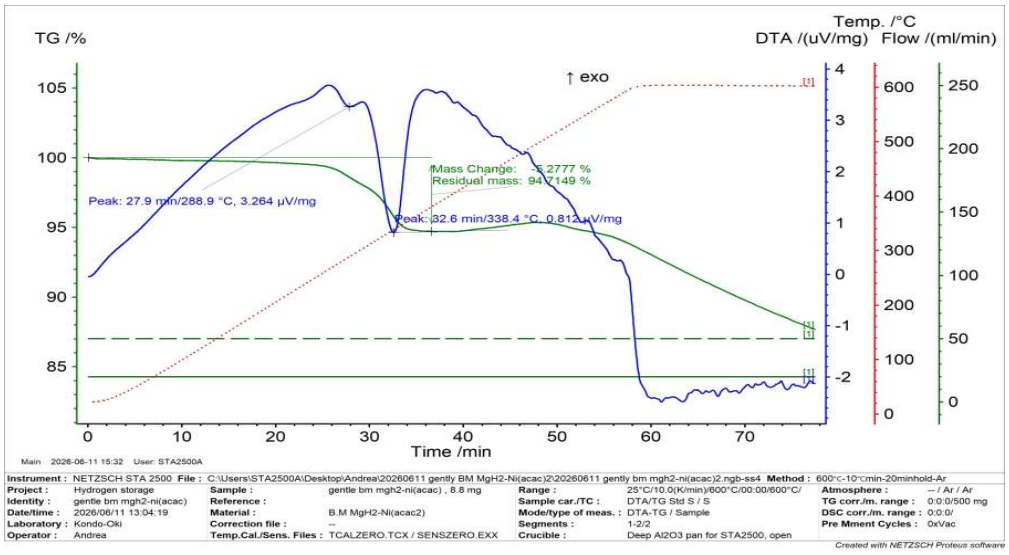
From Figure 30, both MgH_2 + 2wt. % $Ni(acac)_2$ samples have a broad, overlapping peak (marked by a black circle) around 100°C and broad dehydrogenation curve, contrasting with the sharp and relatively narrow peaks of pure ball milled MgH_2 in Figure 29. This may be characteristic of $Ni(acac)_2$ itself. We will investigate and verify whether the broad peak comes from $Ni(acac)_2$ by doing TG-DTA-QMS-TPD of $Ni(acac)_2$ in the future.

The onset temperatures of the gently balled milled sample and the intensively ball milled samples are 257.862°C and 260.643°C, respectively. The intensive ball milled sample first releases the maximum dehydrogenation peak in 288.274°C, and then has a smaller major peak in 329.52°C. This indicates a 2 step reaction process

Meanwhile, the gently ball milled sample has 3 major peaks, in contrast to the 2 major peaks of the intensively ball milled sample. Multi-peak fitting of the TPD curve of the gently ball milled sample is difficult and so far, has to be approximated by many curves. Thus, the peak temperatures of the gently ball milled sample are only an approximation. The first peak is at 284.035°C. The maximum peak dehydrogenation temperature is at the 2nd peak at 327.271°C. As can be seen in Figure 37, this is only a slight decrease from the peak dehydrogenation temperature of ball milled MgH_2 in 338°C. In the gently ball milled sample, there is an extra high temperature, broad peak in the range from 375.61°C-429.59°C with the maximum point in 392.835°C. We theorise that the extra high-temperature peak in the gently ball milled sample may be produced from partial incubation as Mg_2Ni phase is undetected. [32] Because no crystalline Mg_2Ni phase was detected after dehydrogenation, the three-step hydrogen release is unlikely to originate predominantly from sequential Mg_2Ni and MgH_2 decomposition. Instead, the additional high-temperature peak at 392.835°C is tentatively attributed to MgH_2 domains that were insufficiently activated by the $Ni(acac)_2$-derived catalyst, possibly because of heterogeneous catalyst dispersion, limited interfacial contact, or an insufficient density of accessible catalytic site. The inability of $Ni(acac)_2$ to generate a detectable Mg_2Ni phase may also have limited the establishment of an efficient Mg–Ni-mediated hydrogen-transfer pathway. As such, there may be uncatalyzed MgH_2 so that there is a high temperature dehydrogenation peak at around 392.835°C. This is reasonable since from Figure 23, the XRD peak of dehydrogenated gently ball milled MgH_2 + 2wt. % $Ni(acac)_2$ is undetected in MgH2 + 2wt. % Ni(acac)2.

Separate TG result from mass interpolation, TG weight loss result obtained from subtracting 100% from weight change in TG result graph, and DTA result graphs of intensive ball milled MgH_2 + 2wt. % $Ni(acac)_2$ are in Figures 28, 29 and 30, respectively. The weight loss

Figure 29 TG-DTA Result of gently ball milled MgH_2+$Ni(acac)_2$



reaches -7.037%. DTA direction indicates the dehydrogenation reaction is endothermic.

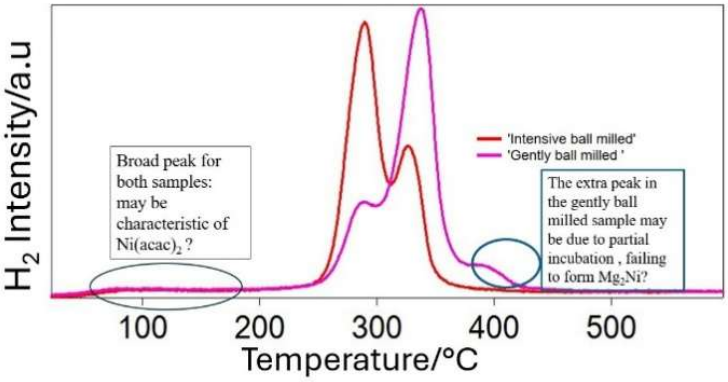


Figure 31 Side-by-side hydrogen release profile comparison of ball-milled MgH2 + 2wt. % Ni(acac)2 sample under different conditions. Intensively ball milled sample is marked by red line while gently ball milled sample is marked by pink line.

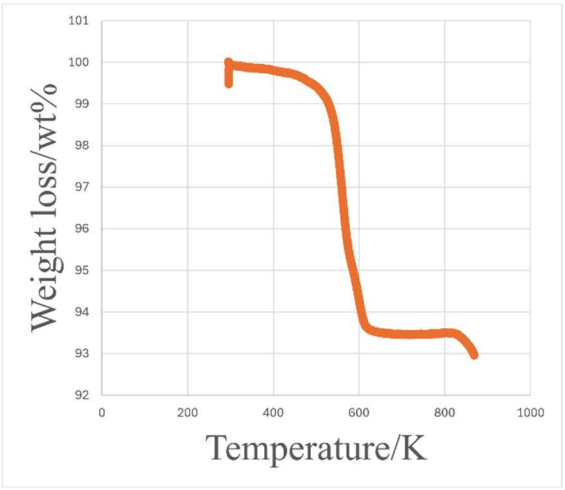


Figure 32 Original TG result of mass change(%) of intensively ball milled MgH_2+$Ni(acac)_2$

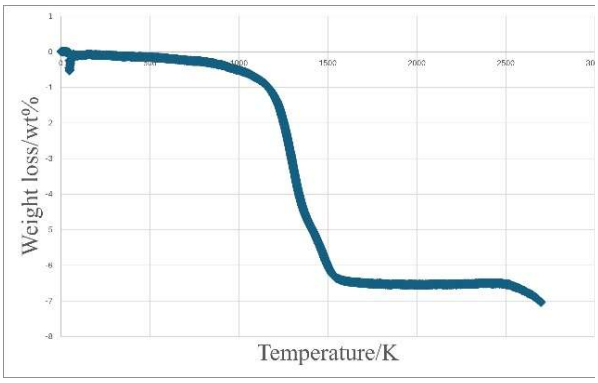


Figure 33 TG weight loss result of intensively ball milled MgH_2+$Ni(acac)_2$

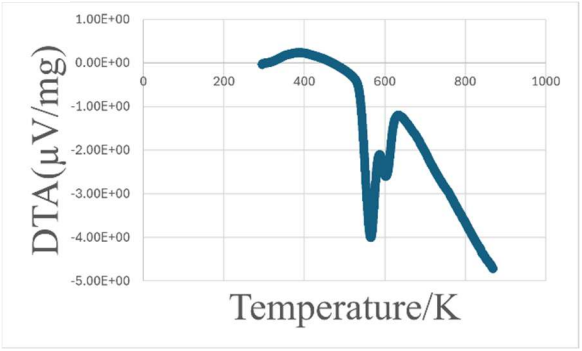


Figure 34 DTA result of intensively ball milled MgH_2+$Ni(acac)_2$

TG result from mass interpolation, TG weight loss result obtained from subtracting 100% from weight change in TG result graph, and DTA result graphs of intensive ball milled MgH_2 + 2wt. % $Ni(acac)_2$ are respectively, are in Figures 31, 32, and 33. respectively. The weight loss reaches -6.1195%. DTA direction indicates the dehydrogenation reaction is endothermic.

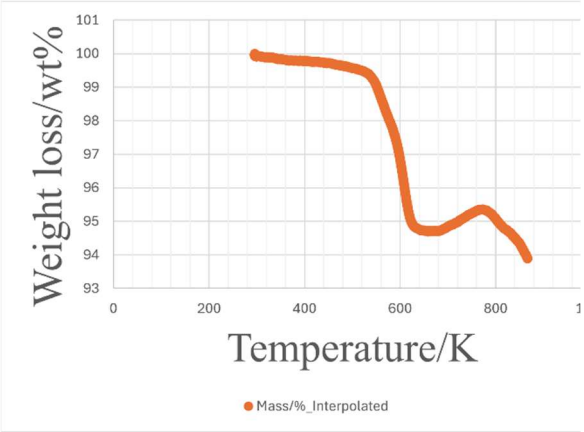


Figure 37 Original TG result of mass interpolated (%) gently ball milled MgH_2 +$Ni(acac)_2$

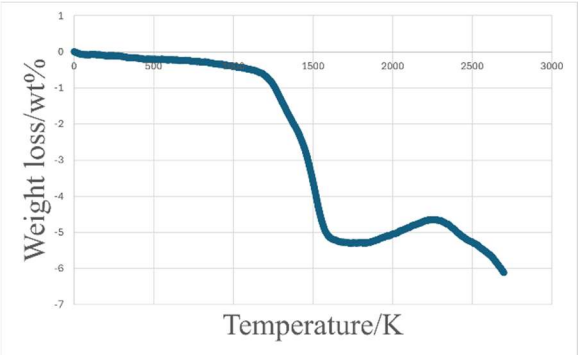


Figure 35 TG weight loss result of gently ball milled MgH_2-$Ni(acac)_2$

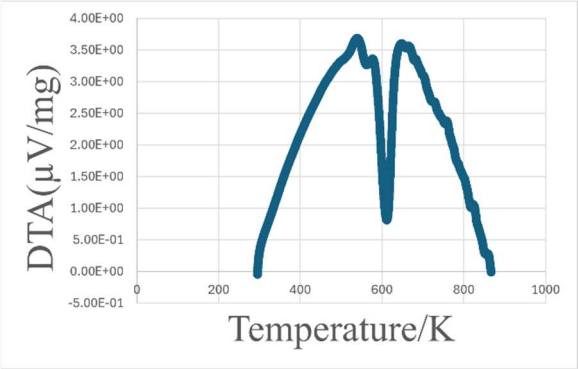


Figure 25 DTA result of gently ball-milled MgH_2+$Ni(acac)_2$

The intensively ball milled MgH_2-$Ni(acac)_2$ has more weight loss compared to the gently ball milled sample upon hydrogenation.

The hydrogen release profiles of ball milled MgH_2 (7 balls), intensively and gently ball milled MgH_2 +2 wt.%$Ni(acac)_2$ are in Figure 34 below.

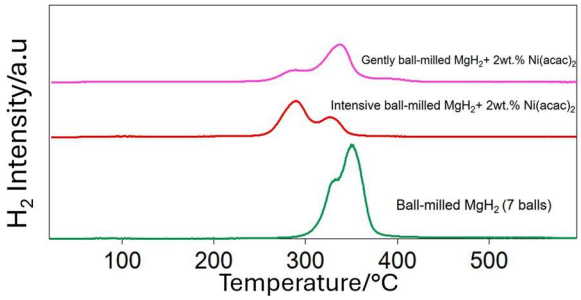


Figure 38 Hydrogen release profile of ball milled MgH_2(7 balls) vs intensively and gently ball-milled MgH_2

From the graph in Figure 38, compared to ball-milled MgH_2, both gently and intensively ball-milled MgH_2 + $Ni(acac)_2$ samples have lower and broader onset dehydrogenation peak. Both intensive and gently ball milled samples have lower peak dehydrogenation temperatures. However, the gently ball milled sample's peak dehydrogenation temperature only decreases slightly compared to pure ball milled MgH_2. One of the factors that can contribute intensive ball milled MgH_2+$Ni(acac)_2$'s better hydrogen release performance might be it decomposes into the hydrogen pump "Mg_2Ni" after dehydrogenation, which help promote dehydrogenation/hydrogenation kinetics.

8 Conclusions and Future Research Plan

In our first step to investigate hydrogen release/storage of ball milled MgH_2 +nickel complexes to improve hydrogen release/storage of MgH_2, we have successfully ball milled MgH_2+$Ni(acac)_2$ using two different methods and conditions: the "intensive" method and the "gentler method" to optimise the conditions for future ball milling of MgH_2 and $Ni(acac)_2$. In comparing the two methods, both have lower dehydrogenation temperatures compared to pure ball milled MgH_2. However, the gently ball milled MgH_2+$Ni(acac)_2$ has only slightly lower peak dehydrogenation temperature than MgH_2. Of all 3 samples, intensively ball milled MgH_2+$Ni(acac)_2$ has the best hydrogen release performance with lowest peak dehydrogenation temperature. The presence of the "hydrogen pump" Mg_2Ni in the intensively ball-milled MgH_2+$Ni(acac)_2$ may help promote dehydrogenation kinetics. Based on this, the intensive ball milling method may be the standard ball milling condition for other nickel

complexes tested in the future.

Our immediate next step is that peak deconvolution needed to determine the exact onset dehydrogenation temperature of gently ball milled MgH_2+$Ni(acac)_2$ samples. We also plan to use Scherrer's equation to calculate the crystalline size of all XRD samples done so far. FT-IR will also be done on ball-milled MgH_2 before dehydrogenation (4 balls) and after dehydrogenation (7 balls), intensively ball-milled MgH_2+$Ni(acac)_2$ samples before and after dehydrogenation, and gently ball-milled MgH_2 before and after dehydrogenation to analyse the bonding of each sample, as in Table 2.

Hopefully, after the optimisation project, we can continue to compare the hydrogen release of ball-milled MgH_2 +different nickel complexes catalyst precursor to investigate the effect of each nickel complexes on the de/hydrogenation kinetics of MgH_2 and elucidate the catalytic reaction pathways. The best performing composite which contains catalyst precursor will then be subsequently evaluated for hydrogen storage using PCT. Finally, graphene or HB will be incorporated into the best-performing MgH_2–Ni composite to further improve hydrogen storage performance and cycling stability. This will path a comprehensive nanocatalyst design guideline into efficient hydrogen release/storage of MgH_2, a promising and low -cost hydrogen storage metal hydride with abundant Mg, and help forming a cleaner, more sustainable future with hydrogen energy source.

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